AISC5000 Foundation of Machine Learning and AI
Lecture 1: Review of Linear Algebra and Calculus

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Logistics

Instructors:

- Prof. JIN Bangti (b.jin@cuhk.edu.hk) Mathematics
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Reference books:

- Bishop, C. M., & Bishop, H. (2023). **Deep Learning: Foundations and Concepts**. Springer Nature.
- Goodfellow, I., Bengio, Y., & Courville, A. (2017). **Deep Learning**. MIT Press.

Course materials: slides and homework available on **Blackboard**.

Lecture structure

Date	Topic	Instructor
Sept. 10	Linear algebra & calculus	Jin
Sept. 17	Probability theory	Zhu
Sept. 24	Chapter 4	Wang
Oct. 1	Holiday	
Oct. 8	Chapter 5	Wang
Oct. 15	Chapter 6	Zhu
Oct. 22	Chapter 7	Jin
Oct. 29	**Midterm**	
Nov. 5	Chapter 8	Wang
Nov. 12	Chapter 9	Jin
Nov. 19	Chapter 10	Zhu
Nov. 26	Chapter 19	Zhu
Dec. 3	Chapter 12	Jin
Dec. 10	**Final exam**	

Assessment scheme

Assessment Type	Percentage
Homework assignments	30%
Midterm, in class	30%
Final exam, final week	40%

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Any related offence will lead to disciplinary action including termination of studies at the University.
Please read:

`http://www.cuhk.edu.hk/policy/academichonesty/`

and avoid any practice that would not be acceptable.

This lecture

Part I: Linear algebra

- vectors and norms
- dot products and geometry
- matrices and linear maps
- bases and linear independence
- inverses and determinants
- eigenvalues and eigenvectors
- orthogonality
- singular value decomposition

Part II: Calculus

- single-variable derivatives
- product, quotient, chain rules
- Taylor series
- integration
- gradients and Hessians
- matrix derivatives

Why review linear algebra and calculus?

Linear algebra in ML

- Data points are vectors.
- Datasets are matrices.
- Neural-network layers are matrix maps.
- PCA uses eigenvectors/SVD.
- Similarity is often measured by inner products.

Calculus in ML

- Training means minimizing a loss function.
- Gradients tell us how to update parameters.
- Hessians describe curvature.
- Taylor expansion explains local approximation.
- Backpropagation is repeated use of the chain rule.

Part I: Linear algebra

Vectors, matrices, geometry, and transformations

Vectors in $\mathbb{R}^n$

An n-dimensional vector $\mathbf{v} \in \mathbb{R}^n$ is an ordered tuple of real numbers:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}.$$

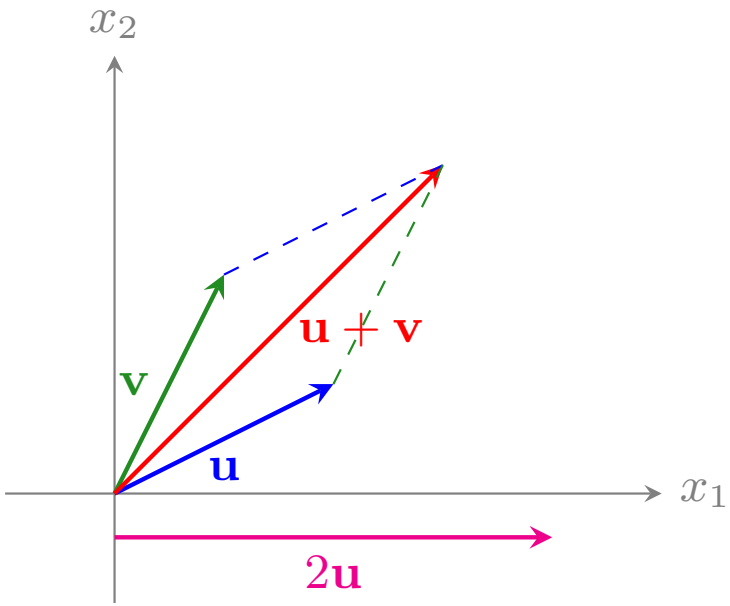
Examples in machine learning

- A grayscale image with 28×28 pixels can be stored as a vector in $\mathbb{R}^{784}$.
- A user profile can be represented by a feature vector.
- Model parameters are often collected into one large vector $\boldsymbol{\theta}$.

Algebraic operations

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}, \qquad c\mathbf{v} = \begin{bmatrix} cv_1 \\ \vdots \\ cv_n \end{bmatrix}.$$



- Addition follows the parallelogram rule.
- Multiplying by c stretches or flips the vector.

Vector norms

For $1 \leq p < \infty$, the ℓ_p norm of $\mathbf{x} \in \mathbb{R}^n$ is

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}.$$

- ℓ_1 norm:

$$\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|.$$

- ℓ_2 norm:

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^{n} x_i^2} = \sqrt{\mathbf{x}^T \mathbf{x}}.$$

- ℓ_∞ norm:

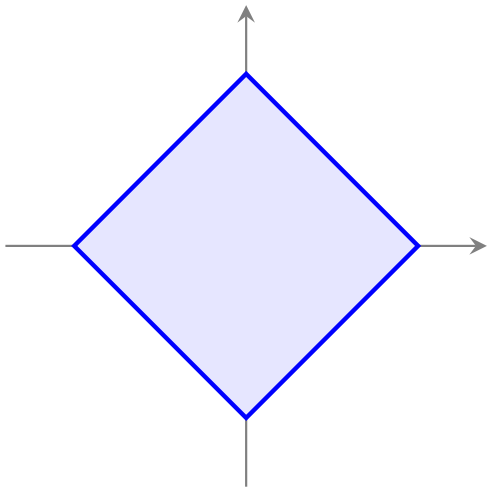
$$\|\mathbf{x}\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

What is a norm?

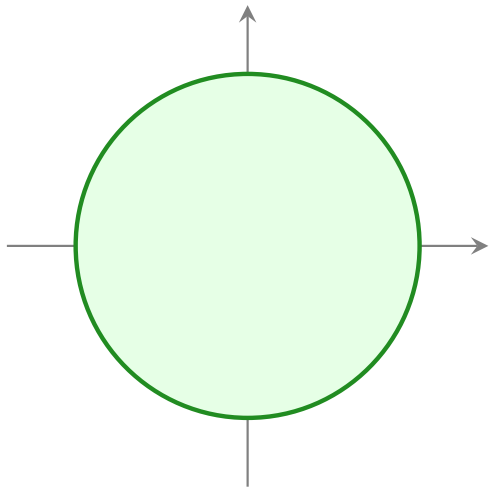
A norm satisfies:

1. $\|\mathbf{x}\| \geq 0$, and $\|\mathbf{x}\| = 0 \iff \mathbf{x} = 0$.
2. $\|c\mathbf{x}\| = |c| \|\mathbf{x}\|$.
3. $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

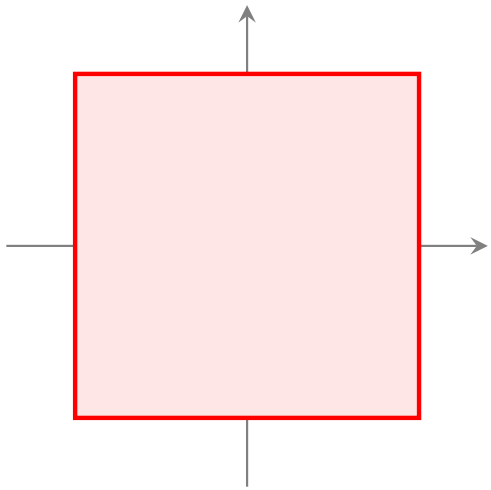
different unit balls



$\|\mathbf{x}\|_1 \leq 1$



$\|\mathbf{x}\|_2 \leq 1$



$\|\mathbf{x}\|_\infty \leq 1$

ML interpretation: Different norms impose different notions of size. For example, ℓ_1-regularization often encourages sparse solutions.

Example: Let

$$\mathbf{x} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}.$$

Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$, $\|\mathbf{x}\|_\infty$

Dot product

The dot product of $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^{n} u_i v_i.$$

Geometrical meaning: $\mathbf{u}^T \mathbf{v} = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \cos\theta$

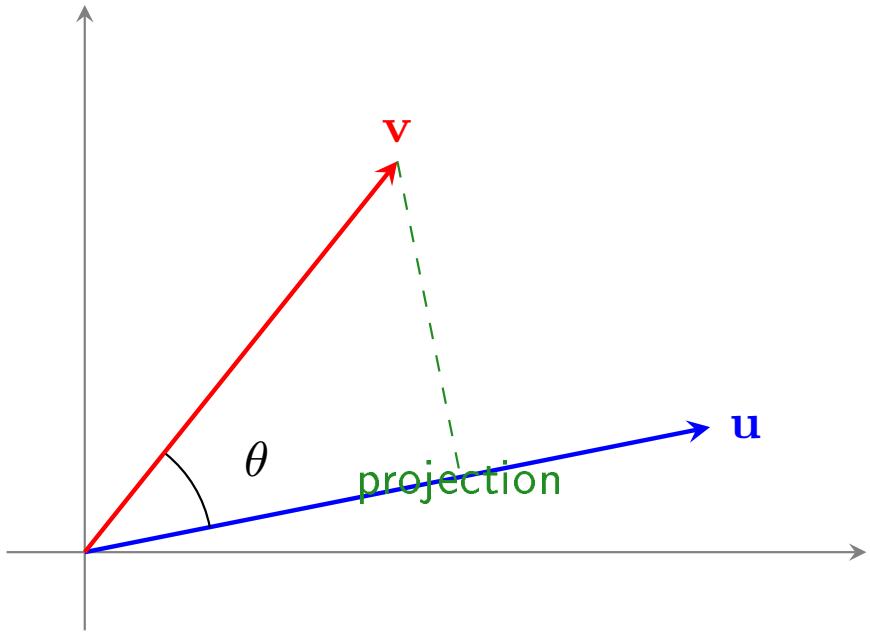
- positive dot product: angle is acute,
- zero dot product: vectors are orthogonal,
- negative dot product: angle is obtuse.

Cauchy–Schwarz inequality

$$|\mathbf{u}^T \mathbf{v}| \leq \|\mathbf{u}\|_2 \|\mathbf{v}\|_2.$$

Equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

dot product and angle



$$\mathbf{u}^T\mathbf{v} = \|\mathbf{u}\| \times \text{length of projection of } \mathbf{v} \text{ onto } \mathbf{u}.$$

Example: Let

$$\mathbf{u} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Find $\cos\theta$.

Then

$$\mathbf{u}^T\mathbf{v} = 3 \cdot 1 + (-4) \cdot 2 = 3 - 8 = -5.$$

Also:

$$\|\mathbf{u}\|_2 = 5, \qquad \|\mathbf{v}\|_2 = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

Therefore

$$\cos\theta = \frac{\mathbf{u}^T\mathbf{v}}{\|\mathbf{u}\|_2\|\mathbf{v}\|_2} = \frac{-5}{5\sqrt{5}} = -\frac{1}{\sqrt{5}}.$$

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Matrices

An $m \times n$ matrix A is an array with m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

In machine learning

- A dataset with m samples and n features is often stored as $X \in \mathbb{R}^{m \times n}$.
- A linear model has the form

$$\hat{\mathbf{y}} = X\mathbf{w}.$$

- A neural-network layer often computes

$$\mathbf{z} = W\mathbf{x} + \mathbf{b}.$$

Matrix taxonomy

- Square matrix: $m = n$.
- Diagonal matrix:

$$D = \text{diag}(d_1, \ldots, d_n).$$

- Identity matrix:

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

- Symmetric matrix:

$$A = A^T.$$

Important examples in ML

- Covariance matrices are symmetric positive semidefinite.
- Kernel matrices are often symmetric positive semidefinite.
- Hessian matrices are symmetric under mild smoothness assumptions.

Matrix operations

For $A, B \in \mathbb{R}^{m\times n}$ and $c \in \mathbb{R}$:

$$(A+B)_{ij} = a_{ij} + b_{ij}, \qquad (cA)_{ij} = ca_{ij}, \qquad (A^T)_{ij} = a_{ji}.$$

For $A \in \mathbb{R}^{m\times k}$ and $B \in \mathbb{R}^{k\times n}$,

$$C = AB \in \mathbb{R}^{m\times n}$$

has entries

$$c_{ij} = \sum_{\ell=1}^{k} a_{i\ell} b_{\ell j}.$$

Dimension rule

$$(m \times k)(k \times n) = m \times n.$$

The inner dimensions must match.

$$(AB)^T = B^T A^T.$$

Example: Compute AB and BA, where

$$A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}.$$

Find AB and BA

$$AB = \begin{bmatrix} 1(2)+3(1) & 1(0)+3(5) \\ -2(2)+4(1) & -2(0)+4(5) \end{bmatrix} = \begin{bmatrix} 5 & 15 \\ 0 & 20 \end{bmatrix}.$$

$$BA = \begin{bmatrix} 2(1)+0(-2) & 2(3)+0(4) \\ 1(1)+5(-2) & 1(3)+5(4) \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ -9 & 23 \end{bmatrix}.$$

Important lesson: Usually, $AB \neq BA$.

Example: Compute AB and BA, where

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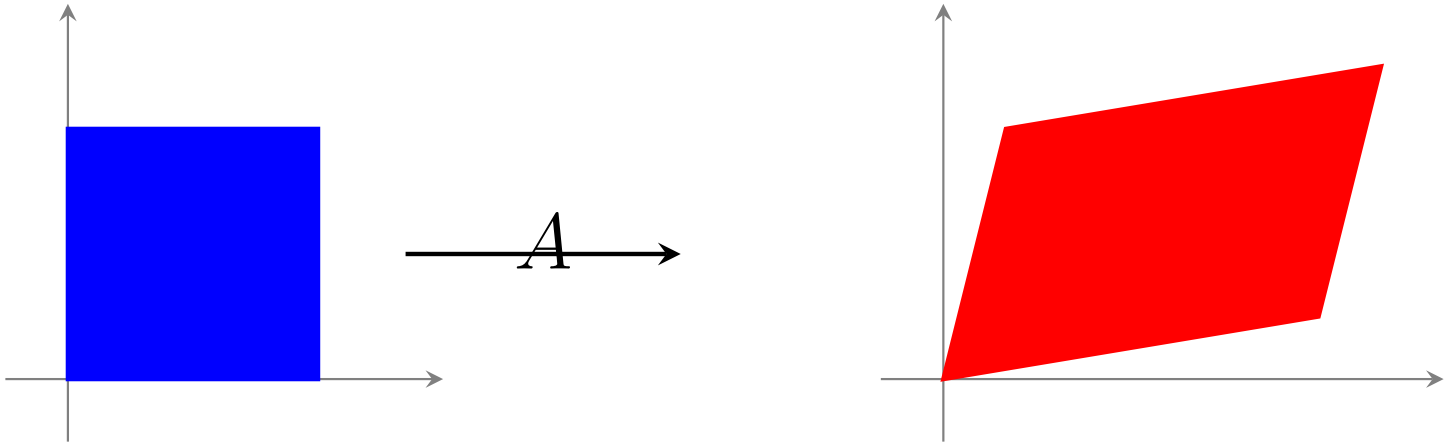
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Important lesson: Usually, $AB \neq BA$.

matrices as transformations: A matrix can rotate, stretch, shrink, reflect, or shear vectors and shapes.



Linear independence, span, and basis

Vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ are linearly independent if

$$c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k = \mathbf{0} \implies c_1 = \cdots = c_k = 0.$$

A set $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_k\}$ is a basis for a subspace V if:

- $\mathcal{B}$ is linearly independent;
- every vector in V can be written as

$$\sum_{i=1}^{k} c_i\mathbf{b}_i.$$

The dimension $\dim(V)$ is the number of vectors in any basis for V.

Example: Are the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

linearly independent?

No. Since

$$\mathbf{v}_2 = 2\mathbf{v}_1,$$

the vectors are linearly dependent.

Are the vectors

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

linearly independent?

Yes. They form the standard basis of $\mathbb{R}^2$.

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Yes. They form the standard basis of $\mathbb{R}^2$.

Systems of linear equations

A linear system has the form

$$A\mathbf{x} = \mathbf{b}.$$

Explicitly:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

$A\mathbf{x} = \mathbf{b}$ has a solution iff $\mathbf{b} \in \mathrm{Col}(A)$.

- Gaussian elimination
- LU factorization
- Cholesky factorization
- conjugate gradient method

Example (least squares): In supervised learning, we often have

$$X\mathbf{w} \approx \mathbf{y}.$$

Usually $X\mathbf{w} = \mathbf{y}$ cannot be solved exactly, so we minimize

$$\min_{\mathbf{w}} \|X\mathbf{w} - \mathbf{y}\|_2^2.$$

If $X^T X$ is invertible, the solution satisfies the normal equation:

$$X^T X\mathbf{w} = X^T \mathbf{y},$$

and hence

$$\mathbf{w} = (X^T X)^{-1} X^T \mathbf{y}.$$

Connection

Linear algebra gives the formula. Calculus explains why this minimizer satisfies the normal equation.

Trace

For $A \in \mathbb{R}^{n \times n}$, the trace is

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} a_{ii}.$$

It is the sum of the diagonal elements.

Useful properties

- $\mathrm{Tr}(A + B) = \mathrm{Tr}(A) + \mathrm{Tr}(B)$.
- $\mathrm{Tr}(cA) = c\,\mathrm{Tr}(A)$.
- $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$, when products are defined.
- $\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB) = \mathrm{Tr}(BCA)$.

Example: Let

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}.$$

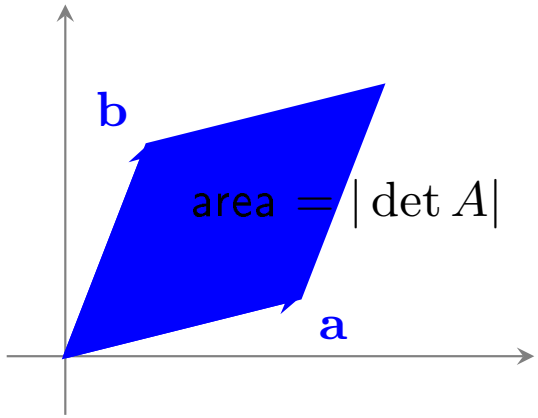
Find $\mathrm{Tr}(A)$

Determinants

For a 2×2 matrix,

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

Geometric meaning: The absolute determinant $|\det(A)|$ is the area scaling factor of the linear transformation $T(\mathbf{x}) = A\mathbf{x}$.



Properties of determinant

- $\det(I_n) = 1$.
- $\det(A^T) = \det(A)$.
- $\det(AB) = \det(A)\det(B)$.
- $\det(cA) = c^n \det(A)$, for $A \in \mathbb{R}^{n\times n}$.
- Matrix determinant lemma:

$$\det(I_n + AB^T) = \det(I_m + A^T B), \quad A, B \in \mathbb{R}^{n\times m}.$$

- Rank-one special case:

$$\det(I_n + \mathbf{a}\mathbf{b}^T) = 1 + \mathbf{a}^T\mathbf{b}.$$

Matrix inverses

A square matrix $A \in \mathbb{R}^{n \times n}$ is invertible if there exists A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n.$$

For an $n \times n$ matrix A, the following are equivalent:

- A is invertible.
- $\det(A) \neq 0$.
- $\operatorname{rank}(A) = n$.
- The columns of A are linearly independent.
- $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{R}^n$.

Example (inverse of a 2×2 matrix): For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

if $ad - bc \neq 0$, then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

Then

$$\det(A) = 2 \cdot 1 - 1 \cdot 1 = 1,$$

so

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Matrix inversion identities

$$(P^{-1} + B^T R^{-1} B)^{-1} B^T R^{-1} = P B^T (B P B^T + R)^{-1}.$$

This is useful when the matrix on the right-hand side is much smaller.

Special case

$$(I + AB)^{-1} A = A(I + BA)^{-1}.$$

Woodbury identity

$$(A + BD^{-1}C)^{-1} = A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1}.$$

These identities appear in Gaussian processes, Bayesian linear regression, and Kalman filtering.

Eigenvalues and eigenvectors

Let $A \in \mathbb{R}^{n \times n}$. A nonzero vector $\mathbf{v} \neq \mathbf{0}$ is an eigenvector with eigenvalue λ if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Equivalently,

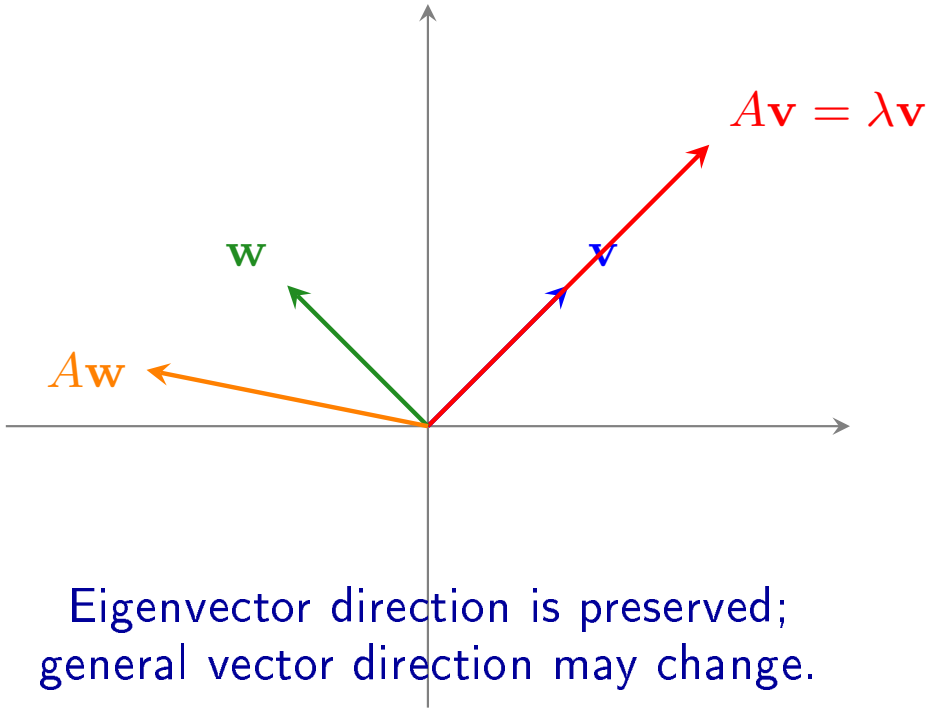
$$(A - \lambda I)\mathbf{v} = 0.$$

Characteristic equation

Nontrivial solutions exist if and only if

$$p(\lambda) = \det(A - \lambda I) = 0.$$

An eigenvector is a direction that the matrix does not rotate. The matrix only stretches, shrinks, or flips it.



Eigenvector direction is preserved;
general vector direction may change.

Example: Find eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

The characteristic polynomial is

$$\det \begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

Thus

$$\lambda^2 - 4\lambda + 3 = 0 \quad \Rightarrow \quad \lambda_1 = 3, \quad \lambda_2 = 1.$$

For $\lambda_1 = 3$:

$$(A - 3I)\mathbf{v} = 0 \quad \Rightarrow \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For $\lambda_2 = 1$:

$$(A - I)\mathbf{v} = 0 \quad \Rightarrow \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

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The characteristic polynomial is

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For $\lambda_2 = 1$:

$$(A - I)\mathbf{v} = 0 \quad \Rightarrow \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Trace and determinant from eigenvalues: For $A \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$,

$$\mathrm{Tr}(A) = \sum_{i=1}^{n} \lambda_i, \qquad \det(A) = \prod_{i=1}^{n} \lambda_i.$$

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

The eigenvalues are 3 and 1.

$$\mathrm{Tr}(A) = 2 + 2 = 4 = 3 + 1,$$
$$\det(A) = 4 - 1 = 3 = 3 \cdot 1.$$

Symmetric matrices

If $A \in \mathbb{R}^{n \times n}$ is real and symmetric, i.e. $A = A^T$, then:

1. all eigenvalues of A are real;
2. eigenvectors corresponding to distinct eigenvalues are orthogonal;
3. A is orthogonally diagonalizable:

$$A = QDQ^T,$$

where $Q^TQ = I_n$.

Positive definite matrices

We write $A \succ 0$ if

$$\mathbf{x}^T A\mathbf{x} > 0 \quad \text{for all} \quad \mathbf{x} \neq 0.$$

Equivalently, all eigenvalues satisfy $\lambda_i > 0$.

Inner product spaces

An inner product $\langle \mathbf{u}, \mathbf{v} \rangle$ on a real vector space V satisfies:

- symmetry:
$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle;$$
- linearity:
$$\langle c\mathbf{u} + \mathbf{w}, \mathbf{v} \rangle = c\langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle;$$
- positive-definiteness:
$$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0, \quad \langle \mathbf{v}, \mathbf{v} \rangle = 0 \iff \mathbf{v} = 0.$$

The induced norm is
$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

Orthogonality

Vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if

$$\langle \mathbf{u}, \mathbf{v} \rangle = 0.$$

A set $\{\mathbf{q}_1, \ldots, \mathbf{q}_k\}$ is orthonormal if

$$\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

If $Q \in \mathbb{R}^{n \times n}$ has orthonormal columns, then

$$Q^T Q = I_n, \qquad Q^{-1} = Q^T.$$

ML connection

Orthogonality gives numerical stability and is central in PCA, QR factorization, and SVD.

Singular value decomposition

Any real matrix $A \in \mathbb{R}^{m \times n}$ can be factored as

$$A = U \Sigma V^T,$$

where:

- $U \in \mathbb{R}^{m \times m}$ is orthogonal;
- $\Sigma \in \mathbb{R}^{m \times n}$ is rectangular diagonal;
- $V \in \mathbb{R}^{n \times n}$ is orthogonal;
- the diagonal entries of Σ are singular values:

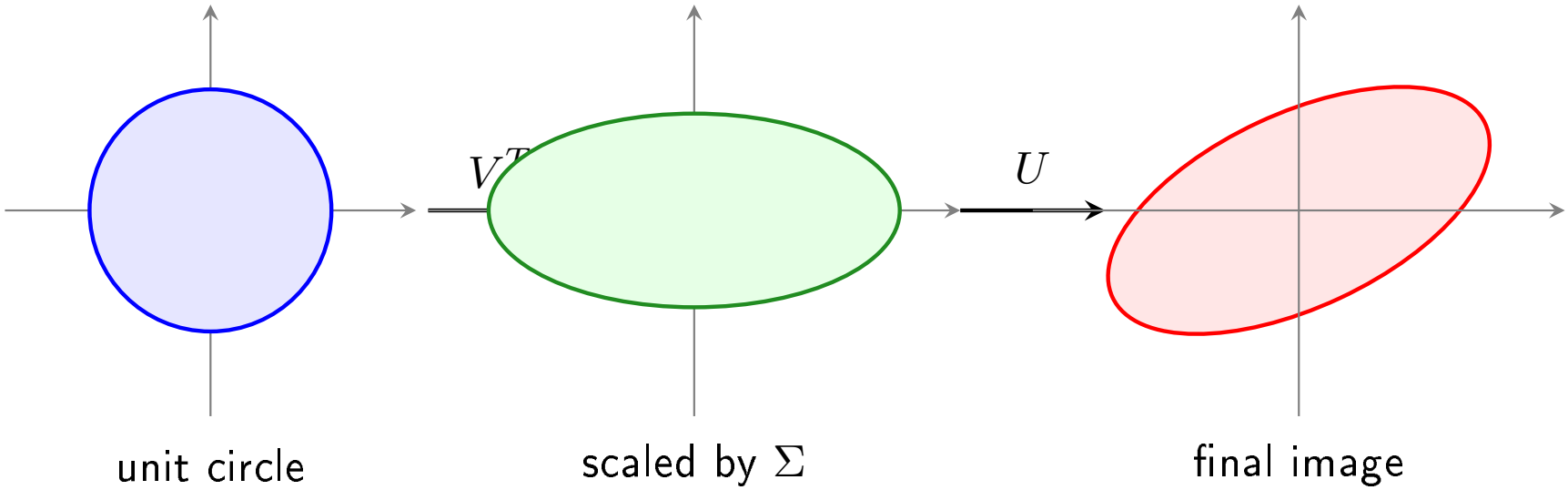
$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$

Geometric view

SVD says every matrix transformation can be decomposed into:

$$\text{rotation/reflection} \quad \rightarrow \quad \text{axis scaling} \quad \rightarrow \quad \text{rotation/reflection}.$$

SVD maps a circle to an ellipse



$$A = U\Sigma V^T.$$

SVD and eigenvalues

Relationships to A^TA and AA^T:

- V comes from diagonalizing A^TA:

$$A^TA\mathbf{v}_i = \sigma_i^2\mathbf{v}_i.$$

- U comes from diagonalizing AA^T:

$$AA^T\mathbf{u}_i = \sigma_i^2\mathbf{u}_i.$$

- Singular values:

$$\sigma_i = \sqrt{\lambda_i(A^TA)}.$$

Also,

$$A\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad 1 \le i \le r.$$

ML application: PCA and low-rank approximation

Given a centered data matrix X, PCA finds directions of largest variance.
Using SVD:

$$X = U\Sigma V^T.$$

The columns of V are principal directions. A rank-k approximation is

$$X_k = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^T.$$

Interpretation

Keeping only the largest singular values gives:

- dimensionality reduction,
- noise reduction,
- compression,
- visualization of high-dimensional data.

Part II: Calculus

Derivatives, integrals, gradients, and optimization

Derivative

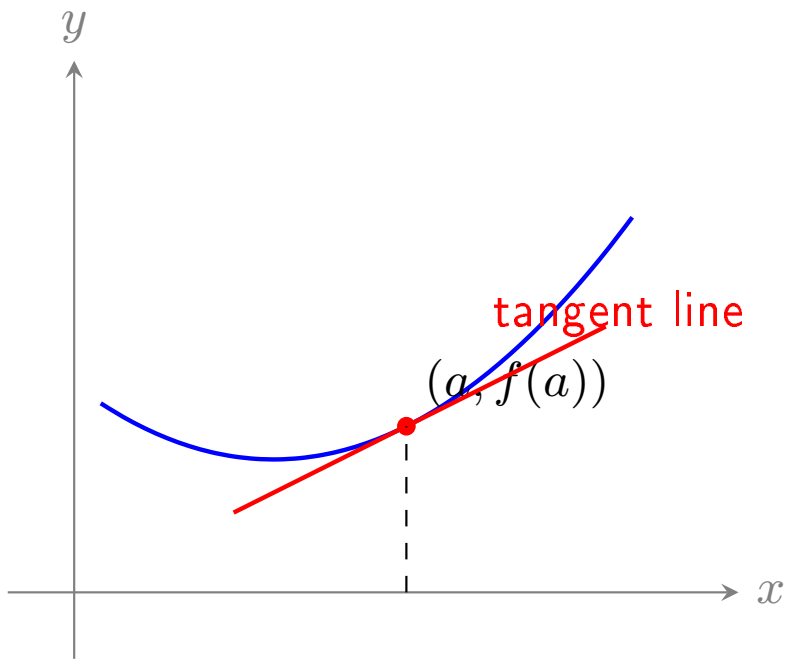
The derivative of $f(x)$ at $x = a$ is

$$f'(a) = \frac{d}{dx} f(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

Meanings

- Geometry: slope of the tangent line to $y = f(x)$ at $(a, f(a))$.
- Physics: instantaneous rate of change.
- Machine learning: sensitivity of loss to parameter changes.

derivative as tangent slope



$f'(a) =$ slope of tangent line at $x = a$.

Example: Find the derivative of

$$f(x) = x^2$$

using the limit definition.

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}.$$

Expanding:

$$f'(x) = \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}.$$

$$= \lim_{h \to 0} \frac{2xh + h^2}{h} = \lim_{h \to 0} (2x + h) = 2x.$$

Thus

$$\frac{d}{dx} x^2 = 2x.$$

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Basic derivative identities

- Power rule:

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

- Exponential:

$$\frac{d}{dx}(e^x) = e^x, \qquad \frac{d}{dx}(a^x) = a^x \ln a.$$

- Logarithm:

$$\frac{d}{dx}(\ln x) = \frac{1}{x}, \quad x > 0.$$

- Linearity:

$$\frac{d}{dx}[cf(x) + g(x)] = cf'(x) + g'(x).$$

Basic derivative rules

- Product rule:

$$\frac{d}{dx}[u(x)v(x)] = u'(x)v(x) + u(x)v'(x).$$

- Quotient rule:

$$\frac{d}{dx}\left[\frac{u(x)}{v(x)}\right] = \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2}.$$

- Chain rule:

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x).$$

ML connection

Backpropagation is essentially repeated application of the chain rule.

Example:

1. Differentiate

$$f(x) = x^3 \sin x.$$

By the product rule:

$$f'(x) = 3x^2 \sin x + x^3 \cos x.$$

2. Differentiate

$$f(x) = \frac{e^x}{x+1}.$$

By the quotient rule:

$$f'(x) = \frac{e^x(x+1) - e^x}{(x+1)^2} = \frac{xe^x}{(x+1)^2}.$$

3. Differentiate

$$y = \ln(x^2+1).$$

By the chain rule:

$$\frac{dy}{dx} = \frac{1}{x^2+1} \cdot 2x = \frac{2x}{x^2+1}.$$

Higher-order derivatives

- First derivative:
$$f'(x).$$
- Second derivative:
$$f''(x) = \frac{d^2 f}{dx^2}.$$
- n-th derivative:
$$f^{(n)}(x) = \frac{d^n f}{dx^n}.$$

Interpretation

- $f'(x)$: slope.
- $f''(x)$: curvature.
- In optimization, curvature tells us whether we are near a minimum or maximum.

Taylor polynomials

The N-th order Taylor polynomial of $f(x)$ centered at $x = a$ is

$$T_N(x) = \sum_{n=0}^{N} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

When $a = 0$, it is called a Maclaurin polynomial.

Taylor's theorem:

$$f(x) = T_N(x) + R_N(x),$$

where

$$R_N(x) = \frac{f^{(N+1)}(c)}{(N+1)!}(x-a)^{N+1}$$

for some c between a and x.

Common Taylor expansions

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots .$$

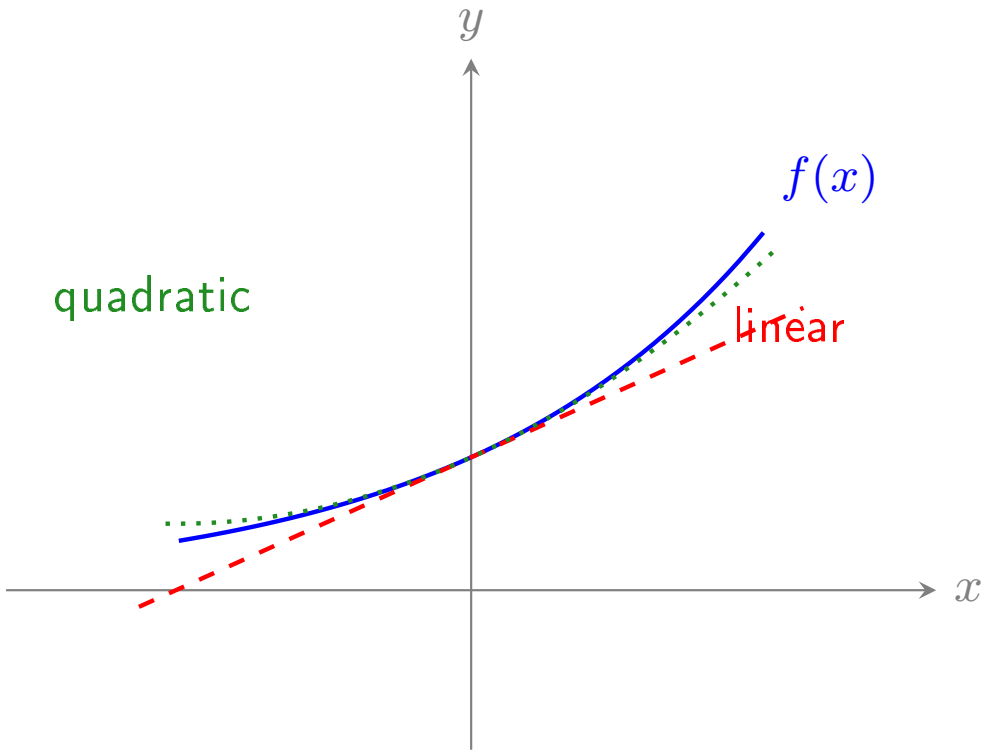
$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots .$$

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots .$$

ML connection

Taylor expansions help analyze local behavior of loss functions.

Taylor approximation



Near the expansion point, low-order polynomials approximate the function well.

Antiderivatives and indefinite integrals

An antiderivative of $f(x)$ is a function $F(x)$ such that

$$F'(x) = f(x).$$

The indefinite integral is

$$\int f(x)\,dx = F(x) + C.$$

Basic integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1.$$

$$\int \frac{1}{x}\,dx = \ln|x| + C.$$

$$\int e^x\,dx = e^x + C, \qquad \int \cos x\,dx = \sin x + C.$$

Definite integral

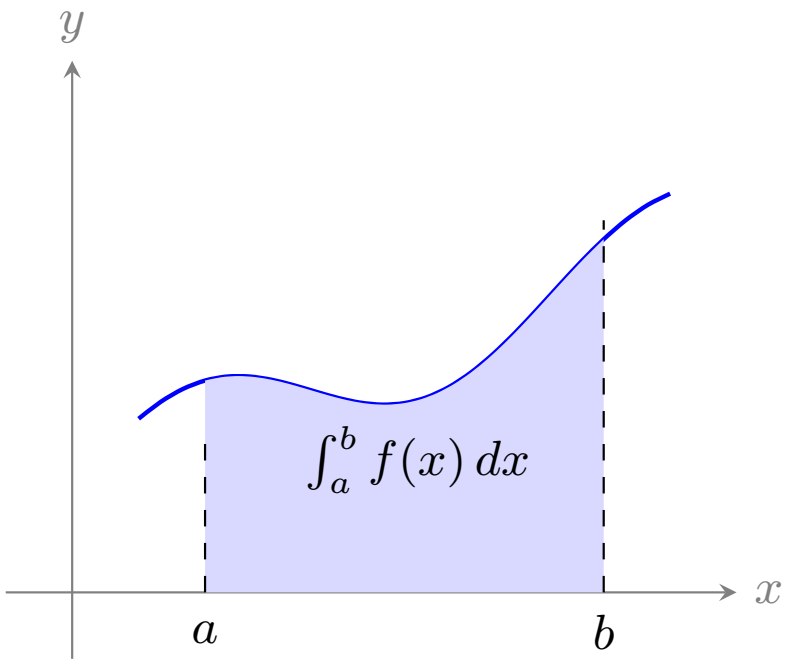
The definite integral over $[a, b]$ is the limit of Riemann sums:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x,$$

where

$$\Delta x = \frac{b-a}{n}, \qquad x_i^* \in [x_{i-1}, x_i].$$

Geometric meaning: For $f(x) \geq 0$, $\int_a^b f(x)\,dx$ is the area under the curve from a to b.



Fundamental theorem of calculus

If f is continuous on $[a, b]$, then

$$g(x) = \int_a^x f(t)\,dt$$

is differentiable and

$$g'(x) = \frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x).$$

If $F'(x) = f(x)$, then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Interpretation

Differentiation and integration are inverse operations.

Multivariable functions

A multivariable function

$$f : D \subseteq \mathbb{R}^n \to \mathbb{R}$$

maps

$$(x_1, \ldots, x_n)$$

to a scalar

$$z = f(x_1, \ldots, x_n).$$

The partial derivative with respect to x_1 is

$$\frac{\partial f}{\partial x_1} = \lim_{h \to 0} \frac{f(x_1 + h, x_2, \ldots, x_n) - f(x_1, x_2, \ldots, x_n)}{h}.$$

Idea

When taking a partial derivative, vary one variable and hold the others fixed.

Gradient

For $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is

$$\nabla f = \begin{bmatrix} f_{x_1} \\ f_{x_2} \\ \vdots \\ f_{x_n} \end{bmatrix}.$$

Geometric meaning

The gradient points in the direction of steepest increase.

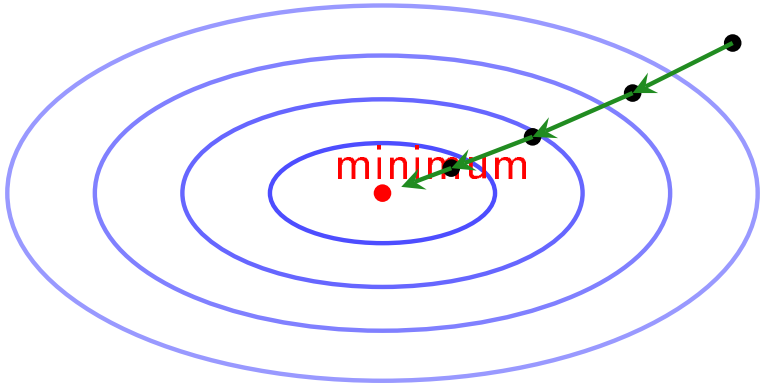
Optimization

To minimize f, gradient descent updates

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k),$$

where $\eta > 0$ is the learning rate.

gradient descent



$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$$

Hessian matrix

The Hessian matrix of $f(x_1, \ldots, x_n)$ is

$$H(\mathbf{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

If the mixed partial derivatives are continuous, H is symmetric.

Hessian and curvature

At a critical point $\mathbf{x}^*$, where

$$\nabla f(\mathbf{x}^*) = 0,$$

the Hessian helps classify the point:

- If $H(\mathbf{x}^*) \succ 0$, local minimum.
- If $H(\mathbf{x}^*) \prec 0$, local maximum.
- If $H(\mathbf{x}^*)$ has positive and negative eigenvalues, saddle point.

ML connection

Deep learning loss landscapes often have many saddle points and flat directions.

Example: Find the gradient and Hessian of

$$f(x,y) = x^3y^2 + \sin(xy).$$

First derivatives:

$$f_x = 3x^2y^2 + y\cos(xy),$$
$$f_y = 2x^3y + x\cos(xy).$$

Therefore

$$\nabla f(x,y) = \begin{bmatrix} 3x^2y^2 + y\cos(xy) \\ 2x^3y + x\cos(xy) \end{bmatrix}.$$

Second derivatives:

$$f_{xx} = 6xy^2 - y^2\sin(xy),$$
$$f_{yy} = 2x^3 - x^2\sin(xy),$$
$$f_{xy} = 6x^2y + \cos(xy) - xy\sin(xy).$$

Thus

$$H(x,y) = \begin{bmatrix} 6xy^2 - y^2\sin(xy) & 6x^2y + \cos(xy) - xy\sin(xy) \\ 6x^2y + \cos(xy) - xy\sin(xy) & 2x^3 - x^2\sin(xy) \end{bmatrix}.$$

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Example: Let

$$f(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}.$$

Then

$$\nabla_{\mathbf{x}} f = (A + A^T)\mathbf{x}.$$

If A is symmetric, then

$$\nabla_{\mathbf{x}} f = 2A\mathbf{x}.$$

The Hessian is

$$H = A + A^T.$$

If A is symmetric:

$$H = 2A.$$

Optimization connection

Quadratic functions are the simplest model for local behavior of smooth loss functions.

Taylor expansion for multivariable functions

Vector linearization:

$$L(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0).$$

Second-order Taylor approximation:

$$Q(\mathbf{x}) = f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0)^T(\mathbf{x} - \mathbf{x}_0) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_0)^T H_f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0).$$

Key message

The gradient gives local slope. The Hessian gives local curvature.

Matrix derivatives

Derivative of inverse

$$\frac{\partial}{\partial x}(A^{-1}) = -A^{-1}\frac{\partial A}{\partial x}A^{-1}.$$

Derivative of log determinant

$$\frac{\partial}{\partial x}\ln|A| = \mathrm{Tr}\left(A^{-1}\frac{\partial A}{\partial x}\right).$$

Trace derivatives

$$\frac{\partial}{\partial A}\mathrm{Tr}(AB) = B^T, \qquad \frac{\partial}{\partial A}\ln|A| = (A^{-1})^T.$$

Mini quiz

1. If

$$\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix},$$

what is $\|\mathbf{x}\|_2$?

2. Are

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

orthogonal?

3. If

$$f(x) = e^{3x},$$

what is $f'(x)$?

4. If

$$f(\mathbf{x}) = \|\mathbf{x}\|_2^2,$$

what is $\nabla f(\mathbf{x})$?

Linear algebra

- Vectors represent data and parameters.
- Norms measure size.
- Dot products measure similarity and angles.
- Matrices represent linear transformations.
- Eigenvalues and SVD reveal important directions.

Calculus

- Derivatives measure local change.
- The chain rule powers backpropagation.
- Gradients drive optimization.
- Hessians describe curvature.
- Taylor expansion connects local geometry and optimization.

References

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- G. H. Golub and C. F. Van Loan, *Matrix Computations*, 3rd ed., Johns Hopkins University Press, 1996.